THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $(x+2)(x+5)$ equals: 1

(a) $x^2+10x+7$ **(b)** x^2+7x+7 **(c)** x^2+10 **(d)** $x^2+7x+10$

2. 8 workers can finish a job in 15 days. 12 workers will finish it in: 1

(a) 10 days **(b)** 11 days **(c)** 22.5 days **(d)** 12 days

3. The marked price of a jacket is ₹ 2500 and it is sold for ₹ 2000. The discount percent is: 1

(a) 22% **(b)** 40% **(c)** 20% **(d)** 18%

4. A triangle has sides 14 cm, 48 cm and 50 cm. It is: 1

(a) right-angled **(b)** equilateral **(c)** not right-angled **(d)** cannot be said

5. In a survey of 60 students, 14 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 168 **(b)** 42 **(c)** 84 **(d)** 94

6. 101×99 equals: 1

(a) 10001 **(b)** 10000 **(c)** 9998 **(d)** 9999

7. A car covers 240 km in 4 hours at a steady speed. In 7 hours it covers: 1

(a) 247 km **(b)** 67 km **(c)** 420 km **(d)** 480 km

8. The price of an article is ₹ 2000 and GST is charged at 5%. The bill amount (in ₹) is: 1

(a) 4200 **(b)** 1890 **(c)** 2100 **(d)** 2310

9. The diagonal of a square of side 3 cm (in cm) is: 1

(a) 6 **(b)** 3 **(c)** $3\sqrt{2}$ **(d)** $\sqrt{19}$

10. ₹ 2400 is divided among A, B and C in the ratio 1:3:4. B's share (in ₹) is: 1

(a) 1200 **(b)** 300 **(c)** 900 **(d)** 800

11. In the expansion of $(2x+3)(4x+5)$ the coefficient of x is: 1

(a) 10 **(b)** 22 **(c)** 23 **(d)** 12

12. ₹ 1200 is divided among A, B and C in the ratio 1 : 3 : 4. The smallest share is: 1

(a) ₹ 240 **(b)** ₹ 150 **(c)** ₹ 450 **(d)** ₹ 600

13. The compound interest on ₹ 5000 for 2 years at 10% per annum compounded annually (in ₹) is: 1

(a) 1000 **(b)** 1155 **(c)** 6050 **(d)** 1050

14. The hypotenuse of a right triangle is 29 cm and one leg is 20 cm. Its area (in cm²) is: 1

(a) 420 **(b)** 230 **(c)** 210 **(d)** 290

15. A car moving at a constant speed covers 210 km in 3 hours. In 8 hours it will cover (km): 1

(a) 218 **(b)** 1680 **(c)** 570 **(d)** 560

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Expand $(2x + 1)(3x + 4)$. 2

17. 9 workers complete a task in 16 days. How many days will 12 workers take? 2

18. A television bought for ₹ 24000 was sold at a loss of 15%. Find the selling price. 2

19. The sides of a triangle are 16 cm, 63 cm, 65 cm. Is it a right triangle? Which angle is the right angle? 2

20. x and y are inversely proportional. When $x = 6$, $y = 20$. Find y when $x = 8$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the amount and the compound interest on ₹ 12500 for 2 years at 8% per annum compounded annually. 3

22. A ladder 29 m long reaches a window 21 m above the ground. Find the distance of the foot of the ladder from the wall. 3

23. A family's monthly expenditure: food 30%, rent 25%, education 20%, others 25%. Find the central angle of each sector of a pie chart. 3

24. Simplify $(x + 2)(x + 3) + (x - 2)(x - 3)$. 3

OR

The ages of two friends are in the ratio 3 : 5 and the sum of their ages is 48 years. Find their ages after 4 years. 3

25. The population of a town was 40000 and became 44000 in a year. Find the percentage growth. 3

OR

The diagonals of a rhombus are 20 cm and 48 cm. Find its side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. ₹ 5250 is divided among A, B and C such that A:B = 2:3 and B:C = 4:5. Find the share of each. 5

OR

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ and verify it for $a = 7$, $b = 3$. 5

27. A bus covers 240 km in 4 hours at a uniform speed. Find 5

(i) its speed

(ii) the distance it covers in 7 hours

(iii) the time it takes to cover 360 km.

OR

A refrigerator has marked price ₹ 30000. The shopkeeper allows a discount of 10% and then charges GST of 18% on the discounted price. Find the amount payable. 5

END OF QUESTION PAPER **Total: 50 Marks**

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